SecureShare

a cloud-based secure file sharing.

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Batch:AWS with python

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Introduction

Cloud computing is a modern computing paradigm that enables users to access computing resources such as servers, storage, databases, and applications over the internet without the need for physical infrastructure. It offers several advantages including scalability, flexibility, cost-effectiveness, and remote accessibility. With the rapid growth of digital data and online services, cloud computing has become an essential technology for developing and deploying applications efficiently. It allows developers to host applications on cloud platforms like AWS, making them accessible from anywhere in the world.

In today's digital environment, secure file sharing is a critical requirement for both individuals and organizations. Traditional file sharing methods often lack proper security mechanisms, leading to risks such as unauthorized access and data leakage. To address these challenges, this project introduces SecureShare, a cloud-based file sharing system that ensures secure and controlled access to files. The system incorporates features such as password protection, expiry-based access, and request-based approval, providing a reliable and user-friendly solution. By deploying the application on a LEMP stack using AWS EC2, the project demonstrates practical implementation of cloud computing concepts in real-world scenarios.

Problem Statement

In the current digital landscape, file sharing is widely practiced through platforms such as email, messaging applications, and cloud storage services. However, these traditional methods often lack adequate security and control mechanisms. Once a file is shared, there is typically no restriction on how long it can be accessed or who can further distribute it, leading to risks such as unauthorized access, data leakage, and misuse of sensitive information.

Additionally, most systems do not provide features like password protection, expiry-based access, or a structured way to regain access after restrictions are applied. This creates a need for a secure and controlled file sharing solution. The SecureShare system is designed to address these challenges by implementing password-protected access, time-based expiry, and a request-approval mechanism, ensuring that file sharing is both secure and manageable in a cloud-based environment.

Objective

To develop a secure file sharing system using cloud computing that ensures safe and reliable data access over the internet.

- To implement password-protected file access so that only authorized users can download the files.
- To provide expiry-based control to limit file availability for a specific time duration.
- To enable request and approval mechanism allowing controlled access after file expiry.
- To deploy the application using LEMP stack on AWS EC2 demonstrating real-world cloud deployment.
- To design a user-friendly interface improving usability and user experience.
- To maintain file metadata in a structured database enabling efficient retrieval and management.
- To handle errors such as wrong passcode or expired files ensuring a smooth and reliable user experience.
- To implement a scalable cloud-based solution that can be extended for larger applications.

Technologies used

Cloud Platform

AWS EC2: Provides a virtual server for hosting the application

Web Server

Nginx: Handles incoming HTTP requests and serves web content efficiently

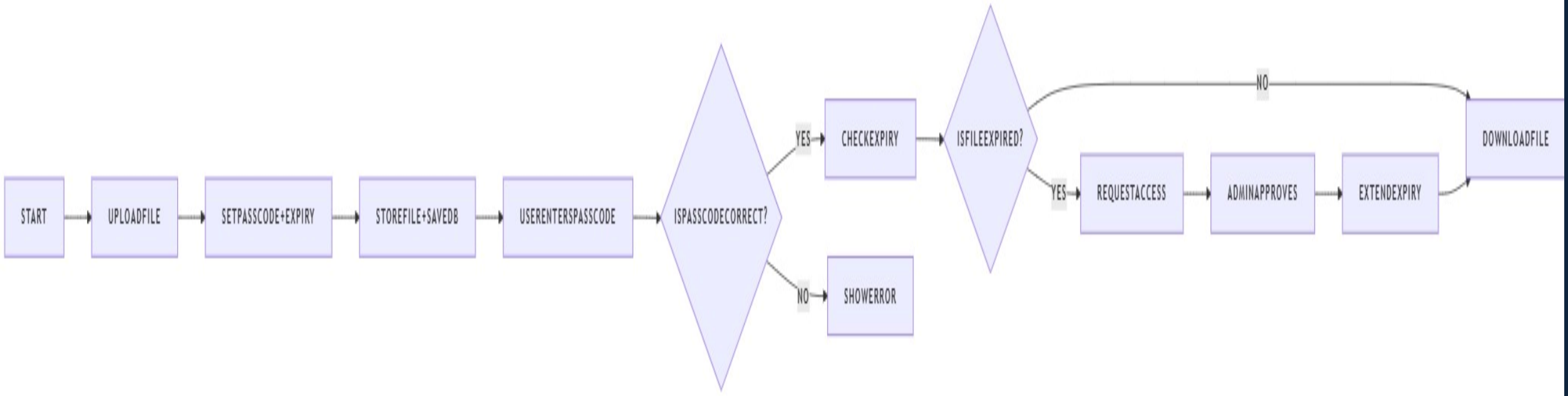
Database

Maria DB: Stores file information including file path, passcode, expiry time, and request status

Backend

PHP: Processes application logic such as file upload, validation, and access control

Project flow

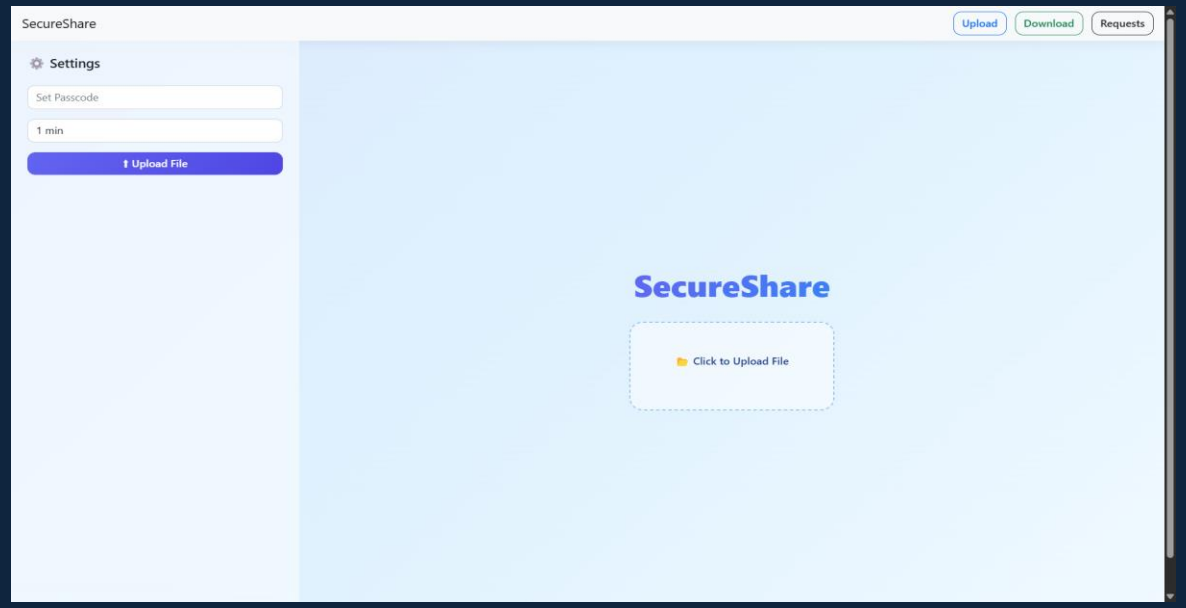


System Architecture

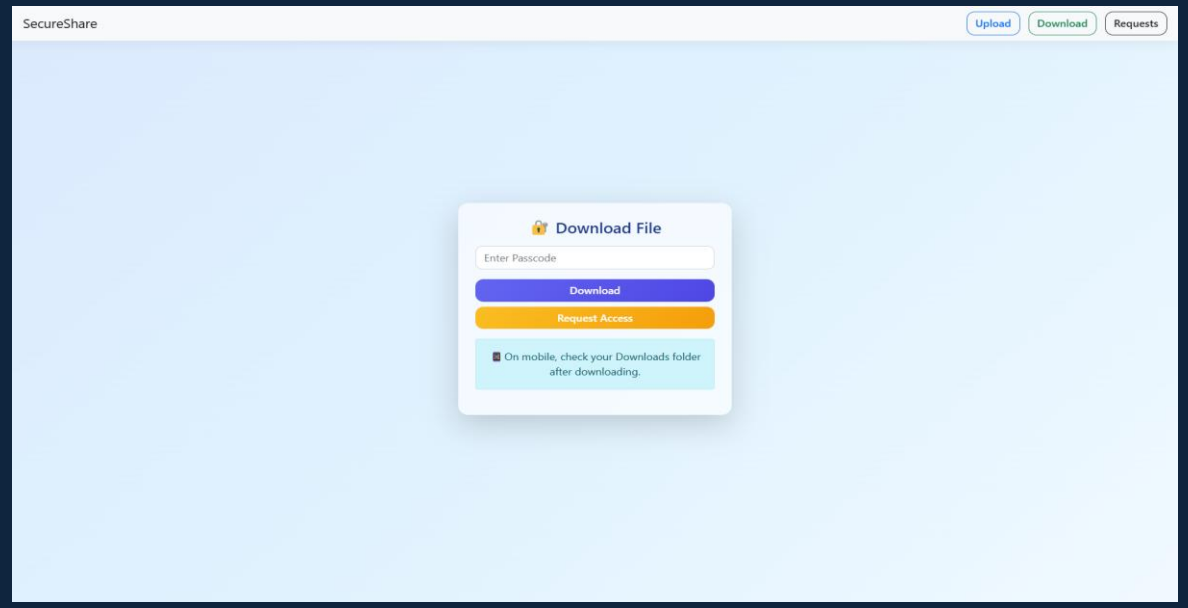
The system architecture of SecureShare is designed around a cloud-based client-server model that supports secure file upload, validation, controlled access, and request-based approval. The user interacts with the system through a web browser interface, where a file is uploaded along with a passcode and expiry time. This request is sent to the cloud server hosted on AWS EC2. The Nginx web server handles the incoming request and forwards it to the PHP backend, which processes the upload logic. The uploaded file is stored in the server directory, while file details such as file name, passcode, file path, and expiry time are stored in the MariaDB database.

When a user attempts to download the file, the system first checks the entered passcode and then verifies whether the file has expired. If the passcode is incorrect, an error message is displayed. If the passcode is correct and the file is still valid, the file is downloaded. If the file has expired, the system provides a request access option. This request is stored in the database and displayed on the admin approval page. Once the admin approves the request, the backend updates the expiry time, allowing the user to download the file again. Thus, the architecture integrates frontend interaction, backend validation, database management, and cloud deployment into a secure and structured workflow.

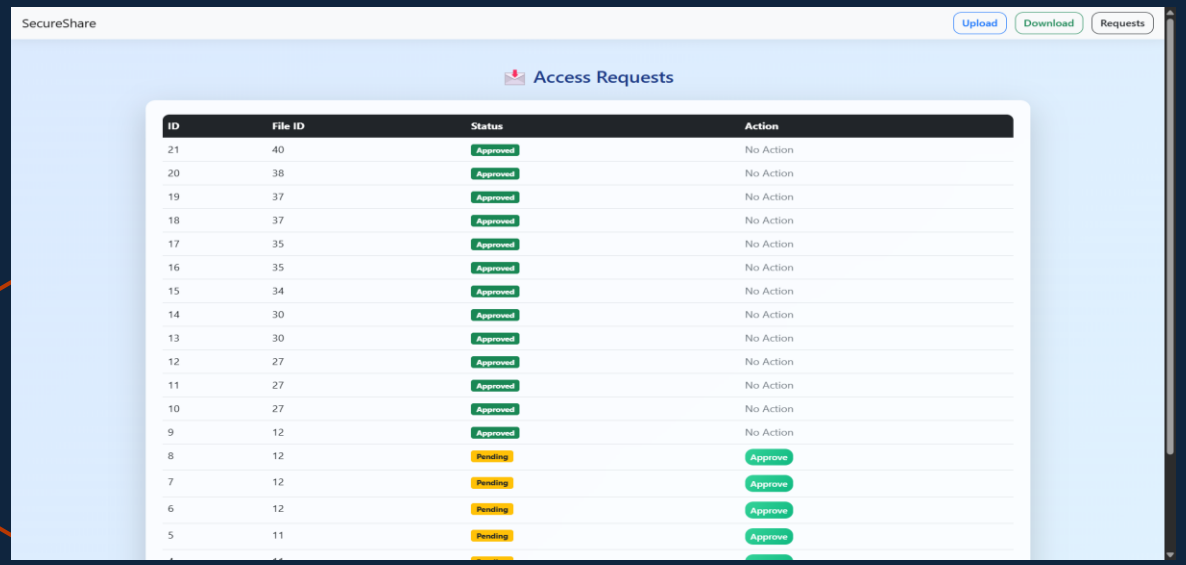
Implementation



The user uploads a file by selecting it and entering a passcode and expiry time. The file is stored on the server and its details are saved in the database

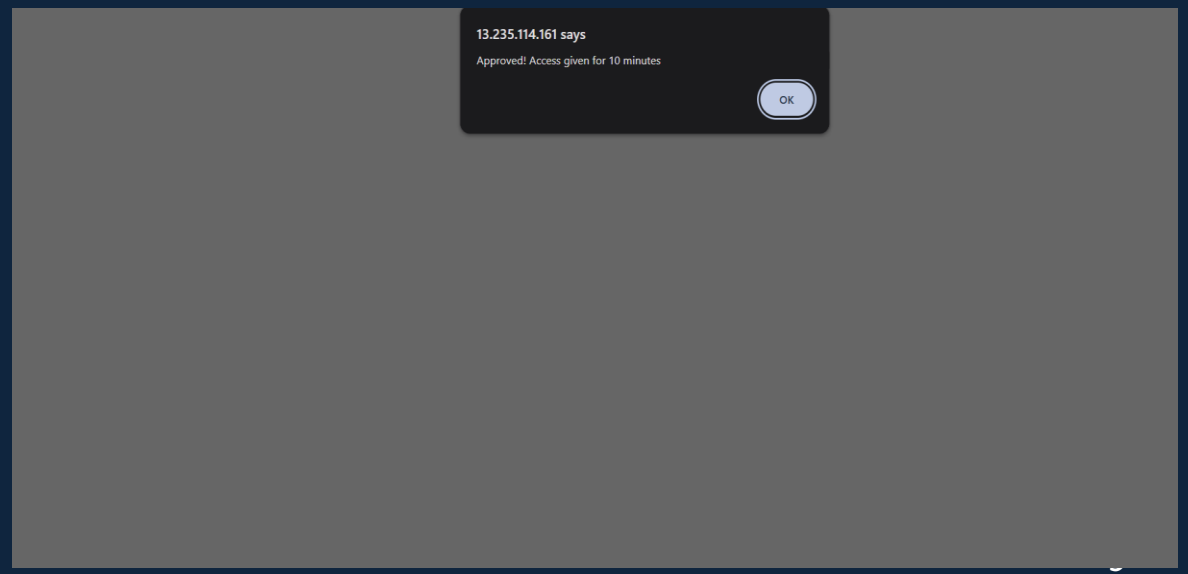


The user enters the passcode to download the file. The system verifies the passcode and allows download if valid



ID	File ID	Status	Action
21	40	Approved	No Action
20	38	Approved	No Action
19	37	Approved	No Action
18	37	Approved	No Action
17	35	Approved	No Action
16	35	Approved	No Action
15	34	Approved	No Action
14	30	Approved	No Action
13	30	Approved	No Action
12	27	Approved	No Action
11	27	Approved	No Action
10	27	Approved	No Action
9	12	Approved	No Action
8	12	Pending	Approve
7	12	Pending	Approve
6	12	Pending	Approve
5	11	Pending	Approve

Users can request access after expiry, which is stored in the database



Admin approves the request, and expiry time is extended.

Result and Conclusion

The SecureShare system was successfully developed and deployed on a cloud platform using a LEMP stack on AWS EC2. All the core functionalities, including file upload, password-protected download, expiry-based access control, and request-based approval, were implemented and tested successfully. The system allows users to securely share files with controlled access, ensuring that only authorized users can download them within a specified time.

The application performed as expected during testing. Users were able to upload files, set passcodes, and define expiry times easily through a user-friendly interface. The download process correctly validated passcodes and restricted access after expiry. The request access feature worked efficiently, allowing users to regain access through admin approval. The integration of frontend, backend, database, and cloud deployment demonstrated a complete working system.

In conclusion, the project effectively showcases the use of cloud computing in building a secure and scalable file sharing application. It highlights the importance of controlled access, data security, and real-world deployment using modern technologies. The system is reliable, easy to use, and can be further enhanced with additional features such as domain integration, authentication, and improved security, making it suitable for practical applications.

Thank you